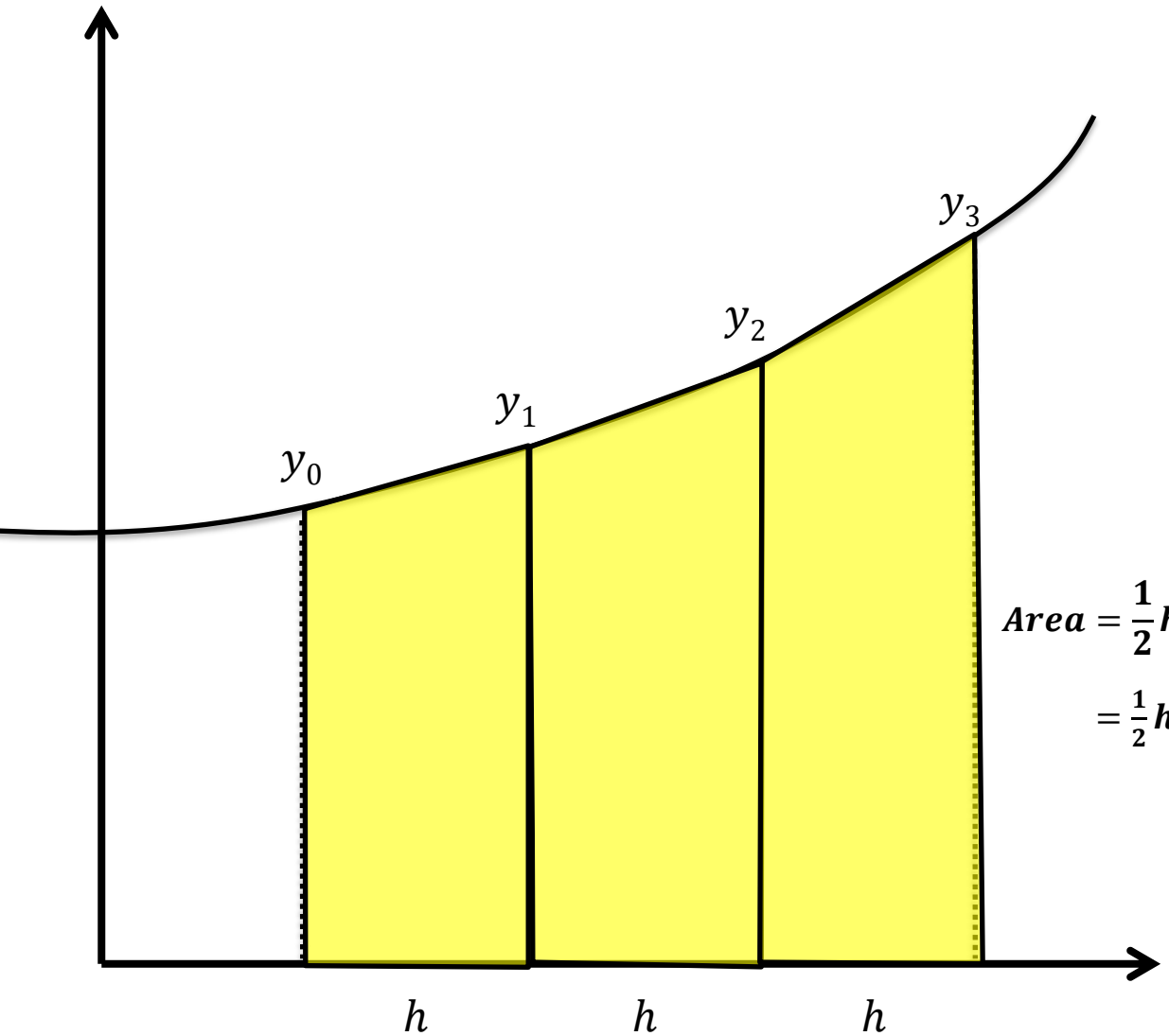
P2 Chapter 11: Integration

The Trapezium Rule

Skill #10: Trapezium Rule



Sometimes finding the exact area under the graph via integration is difficult. Students who have taken GCSE Maths may be familiar with the idea of **approximating the area under a graph by dividing it into trapeziums of equal width**.

$$\boldsymbol{Area = \frac{1}{2}h(y_0 + y_1) + \frac{1}{2}h(y_1 + y_2) + \frac{1}{2}h(y_2 + y_3)}$$

$$\boldsymbol{= \frac{1}{2}h(y_0 \quad + 2(y_1 + y_2) \quad + y_3)}$$

Trapezium Rule

In general:

width of each trapezium

$$\int_a^b y\,dx \approx \frac{h}{2}\left(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\right)$$

Area under curve

is approximately

Example

We're approximating the region bounded between $x = 1$, $x = 3$, the x-axis the curve $y = x^2$, using 4 strips.

Dividing a gap of 2 into 4 strips means each strip will be width 0.5

x	?
y	?

$h =$?

$Area \approx$?

Trapezium Rule

In general:

width of each trapezium

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Area under curve

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Example

We're approximating the region bounded between $x = 1$, $x = 3$, the x-axis the curve $y = x^2$, using 4 strips.

x	1	1.5	2	2.5	3
y	1	2.25	4	6.25	9

Dividing a gap of 2 into 4 strips means each strip will be width 0.5

$h = 0.5$

$$Area \approx \frac{0.5}{2}(1 + 2(2.25 + 4 + 6.25) + 9) = 8.75$$

Trapezium Rule

Edexcel C2 May 2013 (R) Q2

$$y = \frac{x}{\sqrt{(1+x)}}$$

(*a*) Complete the table below with the value of y corresponding to $x = 1.3$, giving your answer to 4 decimal places.

(1)

Fro Tip: You can generate table with Casio calcs . $Mode \rightarrow 3\ (Table)$. Use 'Alpha' button to key in X within the function. Press =

x	1	1.1	1.2	1.3	1.4	1.5
y	0.7071	0.7591	0.8090	?	0.9037	0.9487

(*b*) Use the trapezium rule, with all the values of y in the completed table, to obtain an approximate value for

$$\int_1^{1.5} \frac{x}{\sqrt{(1+x)}}\, dx$$

giving your answer to 3 decimal places.

You must show clearly each stage of your working.

(4)

$\boldsymbol{Area} \approx$?

Trapezium Rule

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(4)

$$\boldsymbol{Area \approx \frac{0.1}{2}(0.7071 + 2(0.7591 + 0.8090 + 0.8571 + 0.9037) + 0.9487) = 0.416}$$

Further Example

Trapezium Rule: $\int_a^b y\,dx \approx \frac{1}{2}h[y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n]$

Q

Given $I = \int_0^{\frac{\pi}{3}} \sec x \;\; dx$

a) Find the exact value of I.

b) Use the trapezium rule with two strips to estimate I.

c) Use the trapezium rule with four strips to find a second estimate of I.

d) Find the percentage error in using each estimate.

a

?

b

?

c

?

d

?

Further Example

Trapezium Rule: $\int_a^b y\,dx \approx \frac{1}{2}h[y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n]$

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a

$$I = [\ln|\sec x + \tan x|]_0^{\frac{\pi}{3}}$$
$$= \ln(2 + \sqrt{3})$$

b

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$
y	1	1.155	2

$$I \approx \frac{1}{2}\frac{\pi}{6}[1 + 2(1.155) + 2] = 1.39$$

c

x	0	$\frac{\pi}{12}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
y	1	1.035	1.155	1.414	2

$$I \approx \frac{1}{2}\frac{\pi}{12}[1 + 2(1.035 + 1.155 + 1.414) + 2] = 1.34$$

d

Error is 5.6% and 1.5%.

Test Your Understanding

Edexcel C4 June 2014(R) Q2

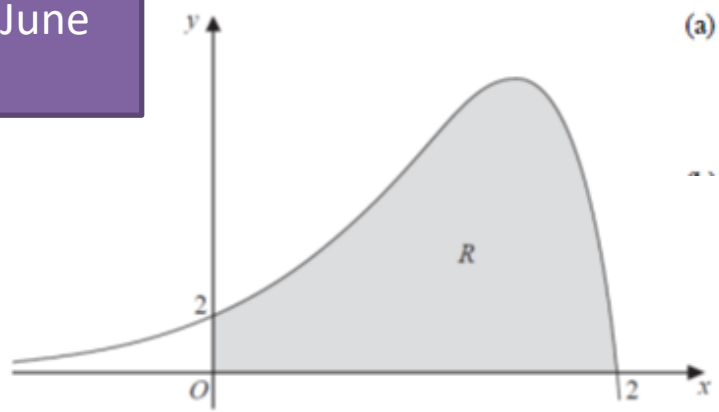


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = (2 - x)e^{2x},$$

The finite region R, shown shaded in Figure 1, is bounded by the curve, the x-axis and the y-axis.

The table below shows corresponding values of x and y for $y = (2 - x)e^{2x}$.

x	0	0.5	1	1.5	2
y	2	4.077	7.389	10.043	0

(a) Use the trapezium rule with all the values of y in the table, to obtain an approximation for the area of R, giving your answer to 2 decimal places.

(3)

(b) Explain how the trapezium rule can be used to give a more accurate approximation for the area of R.

(1)

(c) Use calculus, showing each step in your working, to obtain an exact value for the area of R. Give your answer in its simplest form.

(5)

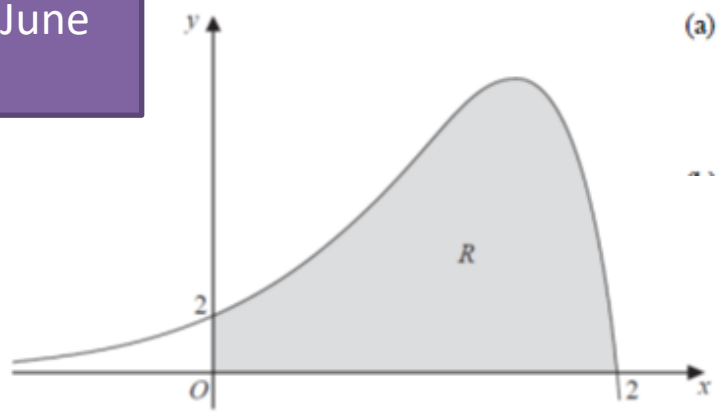
(a) ?

(b) ?

(c) ?

Test Your Understanding

Edexcel C4 June 2014(R) Q2



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(5)

(a) Area $\approx \frac{1}{2} \times 0.5 \times \left[2 + 2(4.077 + 7.389 + 10.043) + 0\right]$

$= \frac{1}{4} \times 45.018 = 11.2545 = 11.25 \text{ (2 dp)}$ — 11.25

B1; M1

A1 cao

[3]

(b) Any one of

- Increase the number of strips
- Use more trapezia
- Make h smaller
- Increase the **number of** x and/or y values used
- Shorter /smaller intervals for x
- More values of y.
- More intervals of x
- Increase n

(c) $\left\{\int (2 - x)e^{2x}\,dx\right\}$, $\left\{\begin{array}{ll} u = 2 - x & \Rightarrow \frac{du}{dx} = -1 \\ \frac{dv}{dx} = e^{2x} & \Rightarrow v = \frac{1}{2}e^{2x} \end{array}\right\}$

$= \frac{1}{2}(2 - x)e^{2x} - \int -\frac{1}{2}e^{2x}\{dx\}$

$= \frac{1}{2}(2 - x)e^{2x} + \frac{1}{4}e^{2x}$

Area $= \left\{\left[\frac{1}{2}(2 - x)e^{2x} + \frac{1}{4}e^{2x}\right]_0^2\right\}$

$= \left(0 + \frac{1}{4}e^4\right) - \left(\frac{1}{2}(2)e^0 + \frac{1}{4}e^0\right)$

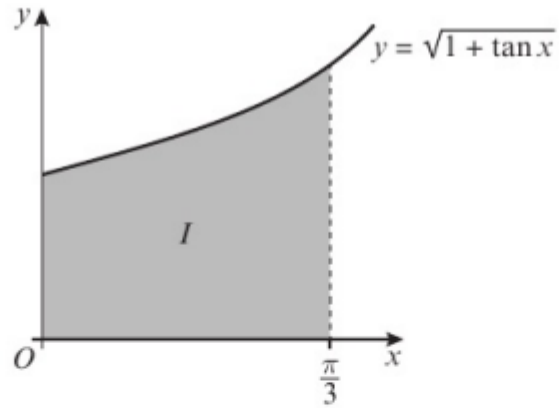
$= \frac{1}{4}e^4 - \frac{5}{4}$

Exercise 11.9

Pearson Pure Mathematics Year 2
Pages 98-100

Homework Exercise

1 The diagram shows a sketch of the curve with equation $y = \sqrt{1 + \tan x}$, $0 \leqslant x \leqslant \frac{\pi}{3}$



a Complete the table with the values for y corresponding to $x = \frac{\pi}{12}$ and $x = \frac{\pi}{4}$ **(1 mark)**

x	0	$\frac{\pi}{12}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
y	1		1.2559		1.6529

Given that $I = \int_0^{\frac{\pi}{3}} \sqrt{1 + \tan x}\, dx$,

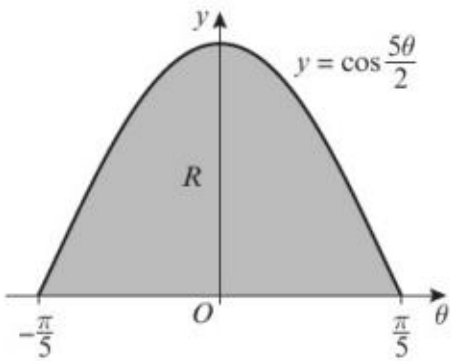
b use the trapezium rule:

i with the values of y at $x = 0$, $\frac{\pi}{6}$ and $\frac{\pi}{3}$ to find an approximate value for I, giving your answer to 4 significant figures; **(3 marks)**

ii with the values of y at $x = 0$, $\frac{\pi}{12}$, $\frac{\pi}{6}$, $\frac{\pi}{4}$ and $\frac{\pi}{3}$ to find an approximate value for I, giving your answer to 4 significant figures. **(3 marks)**

Homework Exercise

2 The diagram shows the region R bounded by the x-axis and the curve with equation $y = \cos\frac{5\theta}{2}$, $-\frac{\pi}{5} \leqslant \theta \leqslant \frac{\pi}{5}$



The table shows corresponding values of θ and y for $y = \cos\frac{5\theta}{2}$

θ	$-\frac{\pi}{5}$	$-\frac{\pi}{10}$	0	$\frac{\pi}{10}$	$\frac{\pi}{5}$
y	0		1		0

a Complete the table giving the missing values for y to 4 decimal places. **(1 mark)**

b Using the trapezium rule, with all the values for y in the completed table, find an approximation for the area of R, giving your answer to 3 decimal places. **(4 marks)**

c State, with a reason, whether your approximation in part **b** is an underestimate or an overestimate. **(1 mark)**

d Use integration to find the exact area of R. **(3 marks)**

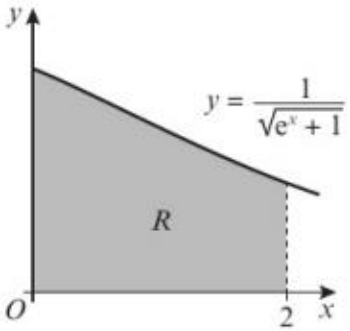
e Calculate the percentage error in your answer in part **b**. **(2 marks)**

3 The diagram shows a sketch of the curve with equation $y = \frac{1}{\sqrt{e^x + 1}}$

The shaded region R is bounded by the curve, the x-axis, the y-axis and the line $x = 2$.

a Complete the table giving values of y to 3 decimal places. **(2 marks)**

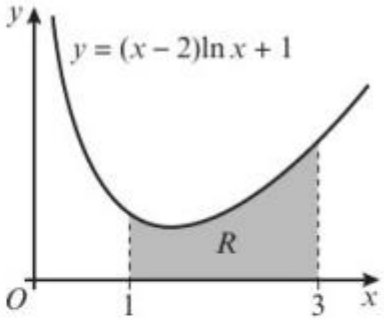
x	0	0.5	1	1.5	2
y	0.707	0.614	0.519		0.345



b Use the trapezium rule, with all the values from your table, to estimate the area of the region R, giving your answer to 2 decimal places. **(4 marks)**

Homework Exercise

4 The diagram shows the curve with equation $y = (x - 2)\ln x + 1$, $x > 0$.



a Complete the table with the values of y corresponding to $x = 2$ and $x = 2.5$. **(1 mark)**

x	1	1.5	2	2.5	3
y	1	0.7973			2.0986

Given that $I = \int_1^3 ((x - 2)\ln x + 1)\,dx$,

b use the trapezium rule

i with values of y at $x = 1$, 2 and 3 to find an approximate value for I, giving your answer to 4 significant figures. **(3 marks)**

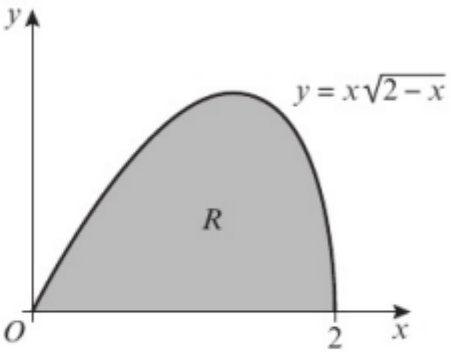
ii with values of y at $x = 1$, 1.5, 2, 2.5 and 3 to find another approximate value for I, giving your answer to 4 significant figures. **(3 marks)**

c Use the diagram to explain why an increase in the number of values improves the accuracy of the approximation. **(1 mark)**

d Show by integration, that the exact value of $\int_1^3 ((x - 2)\ln x + 1)\,dx$ is $-\frac{3}{2}\ln 3 + 4$. **(6 marks)**

Homework Exercise

5 The diagram shows the curve with equation $y = x\sqrt{2-x}$, $0 \leqslant x \leqslant 2$.



a Complete the table with the value of y corresponding to $x = 1.5$. **(1 mark)**

x	0	0.5	1	1.5	2
y	0	0.6124	1		0

Given that $I = \int_0^2 x\sqrt{2-x}\,\mathrm{d}x$,

b use the trapezium rule with four strips to find an approximate value for I, giving your answer to 4 significant figures. **(5 marks)**

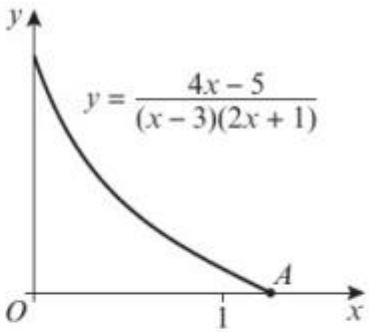
c By using an appropriate substitution, or otherwise, find the exact value of $\int_0^2 x\sqrt{2-x}\,\mathrm{d}x$, leaving your answer in the form $2^q p$, where p and q are rational constants. **(4 marks)**

d Calculate the percentage error of the approximation in part **b**. **(2 marks)**

Homework Exercise

6 The diagram shows part of the curve with equation

$$y = \frac{4x - 5}{(x - 3)(2x + 1)}$$



a Show that the coordinates of point A are $\left(\frac{5}{4}, 0\right)$. **(1 mark)**

b Complete the table with the value of y corresponding to $x = 0.75$. Give your answer to 4 decimal places. **(1 mark)**

x	0	0.25	0.5	0.75	1	1.25
y	1.6667	0.9697	0.6		0.1667	0

Given that $I = \int_0^{\frac{5}{4}} \frac{4x - 5}{(x - 3)(2x + 1)}\,dx$,

c use the trapezium rule with values of y at $x = 0, 0.25, 0.5, 0.75, 1$ and 1.25 to find an approximate value for I, giving your answer to 4 significant figures. **(3 marks)**

d Find the exact value of $\int_0^{\frac{5}{4}} \frac{4x - 5}{(x - 3)(2x + 1)}\,dx$, giving your answer in the form $\ln\left(\frac{a}{b}\right)$. **(4 marks)**

e Calculate the percentage error of the approximation in part **c**. **(2 marks)**

Homework Exercise

7 $I = \int_0^3 e^{\sqrt{2x+1}}\,dx$

a Given that $y = e^{\sqrt{2x+1}}$, complete the table of values of y corresponding to $x = 0.5$, 1 and 1.5. **(2 marks)**

x	0	0.5	1	1.5	2	2.5	3
y	2.7183				9.3565	11.5824	14.0940

b Use the trapezium rule, with all the values of y in the completed table, to obtain an estimate for the original integral, I, giving your answer to 4 significant figures. **(3 marks)**

c Use the substitution $t = \sqrt{2x+1}$ to show that I may be expressed as $\int_a^b kte^t\,dt$, giving the values of the constants a, b and k. **(5 marks)**

d Use integration by parts to evaluate this integral, and hence find the value of I correct to 4 significant figures. **(4 marks)**

Homework Answers

1 **a** 1.1260, 1.4142 **b** **i** 1.352 **ii** 1.341

2 **a** 0.7071, 0.7071 **b** 0.758

c The shape of the graph is concave, so the trapezium lines will underestimate the area.

d 0.8 **e** 5.25%

3 **a** 0.427 **b** 1.04

4 **a** 1, 1.4581 **b** **i** 2.549 **ii** 2.402

c Increasing the number of values decreases the interval. This leads to the approximation more closely following the curve.

d $\int x \ln x \, dx - \int 2 \ln x \, dx + \int 1 \, dx$

$$= \left[\left(\frac{1}{2}x^2 - 2x\right) \ln x - \frac{1}{4}x^2 + 3x\right]_1^3$$

$$= \left(-\frac{3}{2} \ln 3 + \frac{27}{4}\right) - \left(\frac{11}{4}\right) = -\frac{3}{2} \ln 3 + 4$$

5 **a** 1.0607 **b** 1.337 **c** $p = \frac{16}{15}$, $q = \frac{1}{2}$ **d** 11.4%

6 **a** $4x - 5 = 0$, $x = \frac{5}{4}$ **b** 0.3556

c 0.7313 **d** $\ln\left(\frac{49}{24}\right)$ **e** 2.5%

7 **a** 4.1133, 5.6522, 7.3891 **b** 23.25

c $t = \sqrt{2x + 1} \Rightarrow \frac{dt}{dx} = (2x + 1)^{-\frac{1}{2}}$

$\Rightarrow (2x + 1)^{\frac{1}{2}} \, dt = dx \Rightarrow t \, dt = dx$

$\int_0^3 e^{(2x+1)^{1/2}} dx = \int_1^{\sqrt{7}} te^t dt \Rightarrow a = 1, b = \sqrt{7}, k = 1$

d 23.20